Smart E-Library Management System

Centre name:Kandy

Module code: R/618/7483

Student ID:BT/DT24/63

Student name: G. Chamod Madusanka perera

Supervisor's name: MR. Udara

Authenticity declaration form

Student ID: BT/DT24/63

Student Name: G. Chamod Madusanka

Document: Unit 21 Emerging Technology

I certify that the work produced in the mentioned document is my own work and all the referred literature have been properly referenced.

Chamod	2026/07/20
………………………………………	……………………………..
Signature	Date

Abstract

The main objective of this research is to overcome the inefficiency, data entry errors and difficulties in calculating fines in the traditional system based on ledgers and library cards currently used in most libraries in Sri Lanka. As a successful alternative to these problems, instead of technologies such as RFID or IoT, a new system called "Smart E-Library Management System" was created using low-cost QR code technology and web technologies using the Waterfall method.

Here, separate secure logins have been provided to students, library officers and the main administrator, and HTML, CSS, JavaScript, PHP and MySQL technologies have been used to build the system. The most unique feature of this is that there is no need for separate scanners for this and books can be issued and received very easily by scanning QR codes using an ordinary smartphone or webcam. In addition, it includes a real-time database (Live inventory), automatic calculation of fines for overdue books and online reservation facilities.

Black-Box testing of the system confirmed that the system controls borrowing limits very accurately and manual data entry errors are completely avoided. Overall, this system has provided a very practical, efficient and cost-effective solution that minimizes the workload of library staff and saves a lot of time for students.

Acknowledgement

I would like to thank my supervisor **MR. Udara** for his time and support in making my "Smart E-Library Management System" project a success.

I would like to thank the Management and the Board of Directors of **SLTMobitel Nebula Institute of Technology** for their time and resources in training sessions to make this project a success.

I would like to especially thank my best friend who helped me make this project a success. Without his experience and knowledge, it would have taken me much longer to complete this project. Therefore, I would like to thank him again.

Finally, I would like to thank my parents who helped me by donating their time and money to complete this project.

Table content

Table content

1. Introduction

1.1. Background of the Project

Most libraries in Sri Lanka currently use a system that uses ledgers and library cards. The aim of this research is to eliminate this very inefficient system that has been in operation for decades and create a very efficient system based on digital technology. This system can perform tasks such as book inventory management, accurate calculation of fines, efficient recording of book loans and retrievals, and identification of unreturned books and absent students.

1.2. Research Question and Problem Statement

The efficiency of the traditional manual systems is still very low and there can be major errors in lending and receiving books, as well as in collecting fines for late returns. Therefore, the primary research problem of this project is to replace the traditional library management system with a digital technology system and to find out whether the efficiency can be increased by using web technology and QR code scanning technology.

1.3. Project Objectives

The main objective of this project is to create a "Smart e-Library Management System". Its specific objectives are:

- Use QR code scanning technology to issue and receive books
- Create a special user interface for students, library staff, and administrators
- Create a live inventory system for book management
- Automatic calculation of fines for overdue books

1.4. Brief Methodology

I have designed this system as a web-based object-oriented system. I have used **HTML, CSS and JavaScript** to build the front end and **PHP** and **MY SQL** to build the back end. I have used the **HTML5-QRCode Scanner library** to scan the QR codes.

1.5. Importance and Justification of the Project

The creation of this project is very important as it provides direct answers to a large number of existing management problems. Since tasks such as calculating fines are automated, the workload of the staff is greatly reduced, and the project can manage the time of students and the management board due to facilities such as students being able to get their own books by scanning QR codes and knowing their book availability in advance.

2. Literature Review

2.1. Historical Evolution and Background

In the early days, library management was done using handwritten notes, and even today this system is used in many libraries in Sri Lanka. Through this system, activities such as lending and returning books take a lot of time, errors can occur in data entry, and a lot of space is used for storage. Later, technologies such as RFID and Barcode were introduced to avoid this task, but due to the high cost of these technologies and the complexity of technical implementation, their use was abandoned. As a successful alternative, the Denso-Wave Association introduced QR code technology to the world in 1994. QR codes can store a large amount of data compared to a regular barcode.

2.2. Present State

Nowadays, QR code technology has become widespread all over the world, and many modern libraries have adopted it. This means that books can be borrowed and returned through QR code scanning, without the need for a librarian.

2.3. Review of similar studies

Below is a summary of some of the research papers on library systems based on QR technology that I have studied in the past.

Study 1: Web-based QR-code system in library data processing (Khairyzal & Prianto, 2026)

* **Research approach and methodology:** This system was designed for SMA Negeri 1 Bangkinang City using the waterfall methodology, which includes requirements analysis, system design, coding, testing, and maintenance. Black box testing methodology was used to measure its success.

* **Strengths:** The use of QR codes has eliminated human errors that can occur in the process of giving and receiving books

* **Limitations and pitfalls:** The system relies on specific hardware requirements, such as an Intel Core i5 processor and 8GB RAM, to function optimally.

Study 2: Integration of Web-based and Mobile Applications with QR Code (Din & Fazla, 2021)

* **Research Approach and Methodology:** Instead of manually searching for books through the Library of Congress Classification (LCC) system, a system for searching and booking books, including QR codes, via the web and mobile phones, has been introduced.

* **Advantages (Merits):** The system can be implemented at very low cost and users can easily access the data.

* **Limitations (Limitations and Pitfalls):** To read QR codes, the user must have a smartphone with a camera and some technical knowledge. Furthermore, the limitations of the error correction mechanisms (error correction mechanism) of the Reed-Solomon code can cause problems that damage the QR code

Study 3: Library Management System Using RFID and QR Code (Putra et al., 2021)

.

* **Research Approach and Methodology:** This system is developed in .Net C# using both technologies, NFC/RFID cards for member identification and QR codes for book identification.

* **Strengths:** High sensitivity in reading QR code from a distance of 40 cm and even when folded.

* **Limitations and Pitfalls:** Using two technologies (RFID and QR) at the same time increases the complexity and cost of the system. Furthermore, the practical drawback is that the reading of RFID cards is limited to a small range of 7 cm.

Study 4: Smart Library Management System Using QR Code (Singh, 2019)

* **Research Approach and Methodology:** A system has been designed that allows users to independently issue books using a Firebase database and an Android phone camera. The unique approach here is to automatically determine the lending period based on the popularity of the book.

* **Strengths:** It almost completely minimizes the intervention of library staff and makes the user independent and is economically more cost-effective than systems such as IoT.

* **Limitations and Pitfalls:** Since the system is completely operated through the students' personal smart phones, users without phones cannot use the system.

2.4. Approach for the Proposed Research

When analyzing the above studies, it is clear that QR technology is economically and practically suitable for library systems, rather than expensive methods such as RFID and IoT. Therefore, it is more practical to build the system using a systematic software development model such as the waterfall method for the proposed "Web-based QR-code system". To overcome the limitations of other research, a more successful research approach would be to create a web interface that can enter data not only through the user's smartphone, but also through the library's webcam and public computers.

3. Methodology

3.1. Research Design and Theoretical Framework

This project is based on the theoretical framework of Design Science Research (DSR) and the philosophy of "pragmatism". The aim of this project is to identify the problem of inefficiency in libraries and create a solution for it.

Here, the "Waterfall Methodology" was used for software development. Accordingly, the system was designed step by step as follows.

1. First, the interface of the system was created using HTML, CSS, and JavaScript

2. Second, the database was created using MY SQL

3. As a last step, the database and the front end were integrated using PHP to complete the system

3.2.Data Collection and Analysis

Both primary and secondary data methods were used to collect the data required for the system.

* **Primary Data:** Primary data was gathered through personal experiences and direct self-observation as a regular library user. By frequently visiting the library, I was able to practically identify the inefficiencies and difficulties inherent in the traditional manual system. For example, issues faced during administrative processes—such as renewing library cards or sorting delayed borrowing records due to unavoidable absence from the library—served as practical primary insights into the cumbersome nature of the manual ledger system. These personal observations highlighted the urgent need for a digitized, automated system
* **Secondary Data:** Secondary data was collected by reviewing previous research papers and literature related to QR code-based library systems. This helped analyze the limitations of expensive technologies like RFID and justify the selection of a cost-effective and practical QR code solution for the proposed system.

3.3.Project Work Details and Timeline

From November 3, 2025 to July 22, 2026, the development and implementation of the Smart E-Library Management System was carried out systematically, following a structured waterfall methodology. The project was divided into distinct sequential tasks, from initial research to final documentation.

The project development schedule and timeline are illustrated in the Gantt chart below:

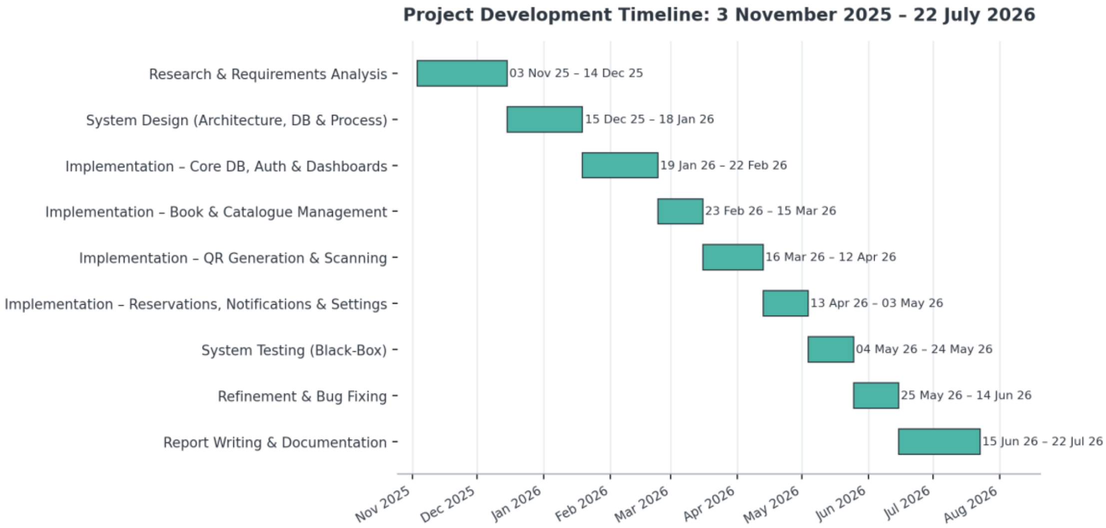

Figure 1: **Project Development Timeline**

Explanation of Project Stages:

Research and Requirements Analysis (03 Nov25 – 14 Dec 25):
Identified operational inefficiencies in traditional library systems, gathered system requirements, reviewed existing literature, and finalized technical frameworks.

System Design (Architecture, Database, and Process) (15 Dec 25 –18 Jan 26):
Designed overall system architecture, database schema (ER diagram), and sequence flow for QR code scanning and user processes.

Implementation – Core DB, Auth & Dashboards (19 Jan 26 – 22 Feb 26): Set up the MySQL database, built user authentication controllers, and developed the base dashboards for students and administrators.

Implementation – Book & Catalogue Management (23 Feb 26 – 15 Mar 26):
Developed the book inventory management modules, category filtering systems, and administrative book addition/removal features.

Implementation – QR Generation & Scanning (16 Mar 26 – 12 Apr 26): Integrated the HTML5-QRCode scanner library and dynamic QR code generation features for self-service book issuing and returns.

Implementation – Reservations, Notifications & Settings (13 Apr 26 – 03 May 26): Built the online book reservation system, automated notification badge counts, fine-calculation logic, and system preference settings.

System Testing (Black-Box) (04 May 26 – 24 May 26): Conducted black-box testing on user logins, QR scans, database operations, and edge cases to ensure system reliability.

Refinement & Bug Fixing (25 May 26 – 14 Jun 26): Resolved minor bugs identified during testing, optimized UI responsiveness, and fine-tuned system performance.

Report Writing & Documentation (15 Jun 26 – 22 Jul 26): Compiled the final project report, including literature reviews, methodology documentation, system architecture diagrams, and references.

3.4. Cost, access and ethical considerations

The three main points to consider when conducting this research and system development are as follows:

- **Cost:**
  Traditionally, RFID or complex IoT systems used for library automation are very expensive. However, in this project, QR code technology was used as a successful alternative. The software was created using completely open source technologies such as PHP, MySQL and HTML5-QRCode Scanner, which kept the software cost to a minimum. Furthermore, the cost of purchasing additional hardware was also avoided, as students could scan QR codes and retrieve books via their own smartphones.
- **Access**
  I used my legitimate access as a registered member of the library to obtain the primary data required by the system and to monitor issues. Also, during the system testing, access control to the system was ensured through authentication methods so that unauthorized persons could not access the database.
- **Ethical Issues**
  The basic understanding required for designing this system was gained based on my personal experience and observations. As a regular library user, I observed the practical difficulties and shortcomings of the book issuance and retrieval process and its efficiency. No special permission was obtained or data was collected from any external party or library staff, and only general observations obtained as a user were used for the initial studies of the project.

3.5.Requirements Analysis

Based on the primary users of the system—students, library officers, and the head administrator—the system requirements were categorized into functional and non-functional aspects.

Functional Requirements

The core features identified and implemented to automate the library processes include:

- **Role-Based Authentication:** Secure, two-step, role-based login (for students, officers, and administrators) supported by robust server-side session handling.

- **Student Registration & Approval:** A self-registration portal for students allowing them to upload identification documents. Accounts remain in a "Pending" state until reviewed and approved by an administrator.

- **Book Catalogue Management:** Administrative tools to add, view, and remove books, featuring the automatic generation of unique book IDs and corresponding QR codes for every title.

- **Catalogue Browsing & Search:** A student-facing interface equipped with live search functionality and category-based filtering.

- **Unified Borrowing Limit (System Constraint):** A strict cross-validation mechanism ensuring a student can hold a maximum of only two concurrent books in total. The system automatically synchronizes online reservations and physical checkouts; if the two-book limit is reached via one method, any further borrowing requests through the other method are automatically blocked.

- **Online Book Reservations:** A reservation system allowing online bookings (subject to the unified limit), managed by an automatic 24-hour collection window that expires both the reservation and the book's "Reserved" status if unclaimed.

- **QR-Based Self-Checkout:** A feature enabling students to independently issue books physically using either a live device camera scan or by uploading an image of the QR code.

- **QR-Based Return Processing:** A staff-side tool to process book returns via QR scanning, with late fines automatically calculated based on the item's due date.

- **Real-Time Countdowns:** Live countdown timers displaying the remaining time for active borrowings and reservations, visible to both students and staff.

- **Notification System:** A dynamic alert system featuring role-specific, section-based unread notification badges for new registrations, reservations, and active borrowings.

- **System Preferences & Backup:** Administrative controls to configure borrowing periods and fine rates, alongside a one-click database backup download option.

- **Advanced Administrative Control:** Exclusive privileges for the head administrator to manage officer accounts, including account deletion and ownership transfers.

Non-functional Requirements

The main non-functional focus of the system was accessibility and cost-effectiveness:

* Browser-based accessibility: The client interface is designed to run seamlessly in any standard web browser using a device camera or an uploaded image for QR scanning, completely eliminating the need for expensive, dedicated reader hardware.

* Cost-effective architecture: The server side is built on a standard Apache, PHP, and MySQL stack (compatible with native environments such as XAMPP or WAMP). This ensures low development and deployment costs, directly addressing the hardware cost limitations found in hybrid RFID approaches noted by Putra, Mahayana, and Vikaxonno (2021).

3.6.System Design

System Architecture

This smart e-library management system has been developed based on a 3-tier architecture model. Three main areas are interconnected:

1.Frontend / User Interface

- The user interface is designed using HTML, CSS, and JavaScript.
- Role-based dashboards are separated into the Student Dashboard for students and the Admin/Main Dashboard for administrators.
- Furthermore, QR code scanning facility (via camera or image upload) has been implemented in the front-end using the html5-qrcode library for book issuance and return.

2.Backend / Server Side:

- Apache, PHP, and MySQL environments are used as the server.
- To separate concerns and maintain a clean architecture, the back-end operations are divided into several specialized controllers: auth_controller.php, user_controller.php, library_controller.php, and system_controller.php. These controllers are implemented to receive requests sent from the front-end via the Fetch API.
- These controllers securely process user requests, execute the SQL queries required to communicate with the database, and return the results as JSON responses to the front-end UI.

3.Database Layer:

MySQL database (smart_e_library_new.sql) is used to store data. This architecture communicates with the database through SQL queries executed through the backend controller and returns the results as JSON responses to the frontend.

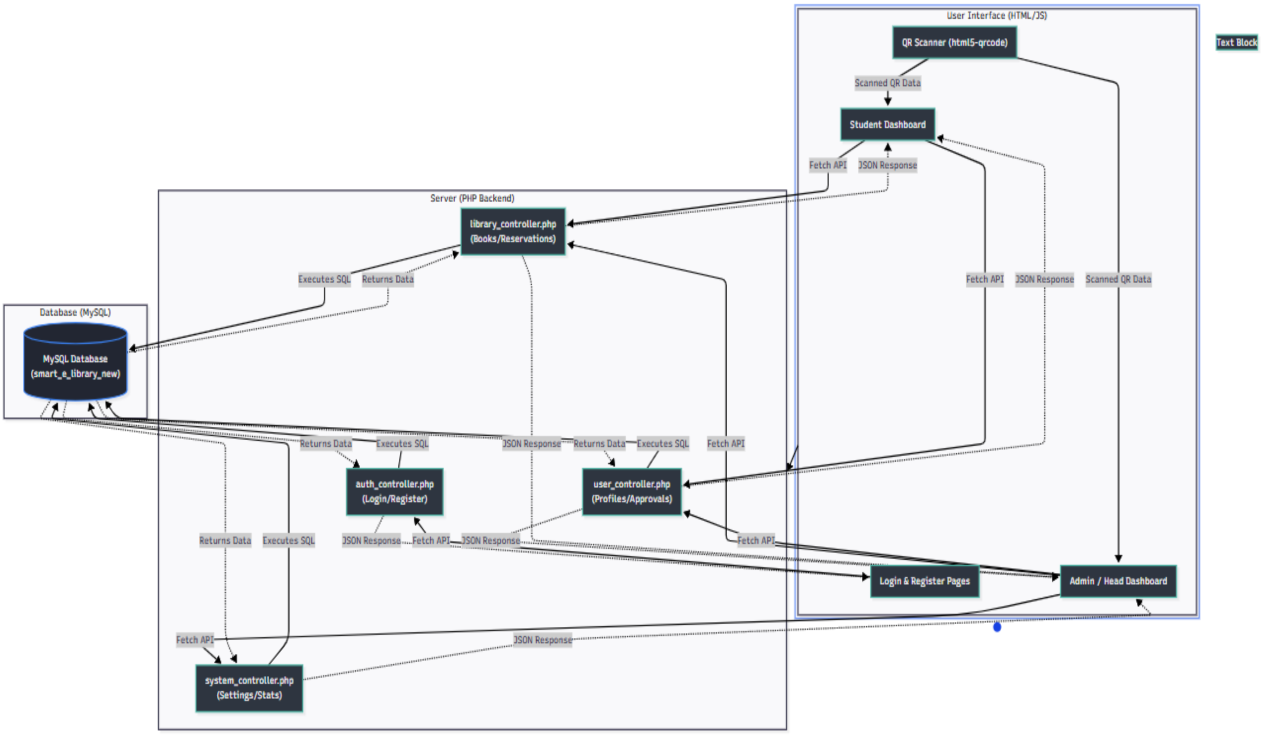

Figure 2: **System Architecture of the Smart E-Library Management System**

Database Design and Structure

For the Smart E-Library system, the database has been created using **MySQL as a relational database consisting of eight interconnected tables.**

Below is a breakdown of the tables and their specific roles in the system:

Admin Table: This table is designed for staff and the primary key is work_id. It includes fields like first_name, last_name, email, and role to properly differentiate between a head librarian and a standard library officer.

Student table: Used to manage library members. The primary key is student_id. In addition to basic contact information, it stores a proof_doc field for account verification purposes and a status field to track whether an administrator has approved the student's registration.

Books table: This acts as the library's basic inventory and uses book_id as its primary key. Important fields include category and status, which are automatically updated to show whether a book is currently available, reserved, or on loan.

Borrowings table: This table is used for the lending process. The Students and Books tables are linked to this table. It uses foreign keys and tracks the issue_date, due_date, and any fine calculated if the student returns the book late.

Reservations table: When a user reserves a title through the online system, the data is saved here. It records the request_date and the current status of the reservation until the book is **collected**.

scan_requests table: This handles the QR code scanning functionality. Every time a book is scanned for quick issuing or returning, it records the request_time and status to process the transaction smoothly without manual data entry.

Notifications table: Added to manage real-time system alerts. It stores notification messages (message) and their read status (**is_read**), ensuring that administrators are immediately notified of new student registrations or pending requests.

system_settings table: A configuration table used to manage global library rules. Without modifying the back-end source code, administrators can update the borrowing_period (the maximum number of days a book is allowed to be kept) and **late_fine** directly from the dashboard.

login_attempts table: This table is newly introduced to enhance system security by tracking failed login attempts, which helps in preventing brute-force attacks on user accounts and securing the authentication process.

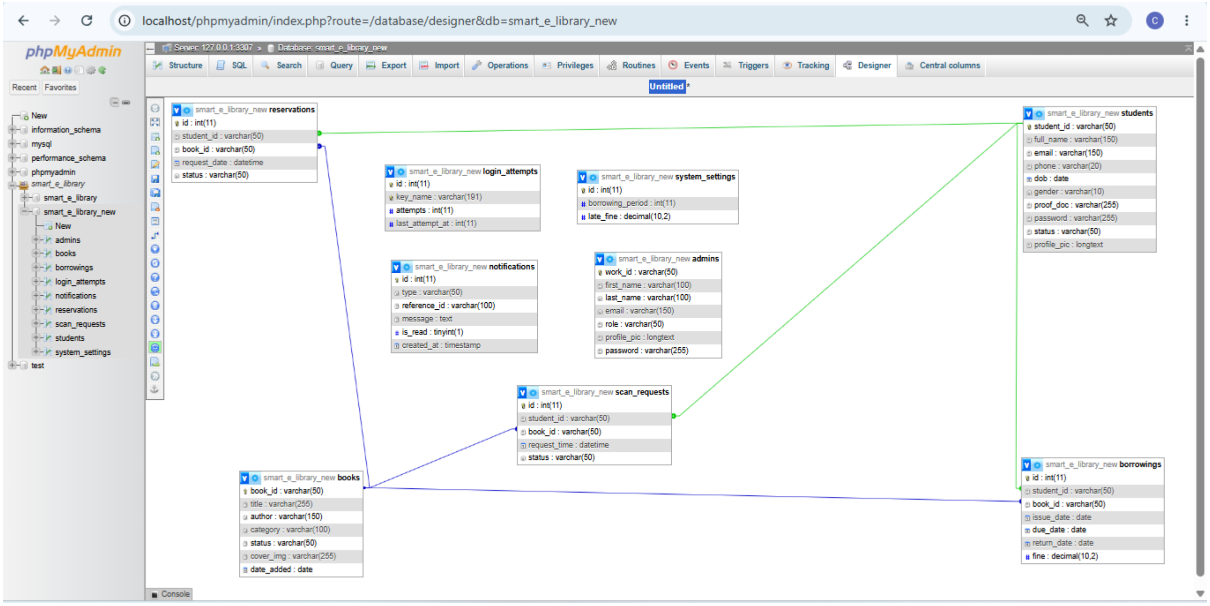

Figure 3: **Database Design and Structure**

Process Design

The Qr code book-issuing sequence

Pitcher4 traces the sequence of interactions that occur when a student checks out a book using a QR scanner. Once the student starts the scanner and presents the QR code of a book, the browser decodes the encoded book_id and sends it to the PHP backend, which checks whether the book exists and its details and returns it for display in a confirmation dialog. Only after the student confirms checkout does the browser send a second request, which actually creates the borrowing record and updates the book's status in the database.

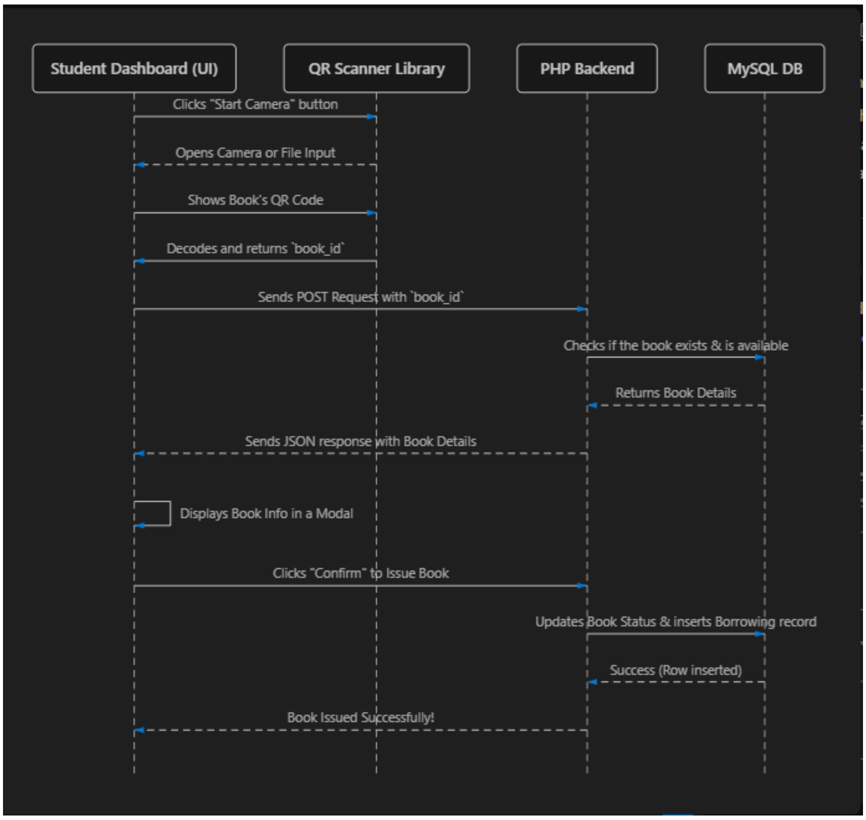

Figure 4: **Sequence Diagram of the QR Code Book-Issuing Process**

The implemented workflow follows this sequence, with one additional safeguard not shown in the diagram: because the confirmation step gives a short window between the initial availability check and the student's final confirmation, the server repeats the availability check a second time when the borrowing request is actually submitted, so that a book cannot be issued twice if two students happen to scan the same title at nearly the same moment.

3.7. Implementation Tools and Technologies

lists the tools and technologies used to build the system.

Component	Technology Used
Front-end markup and styling	HTML5, CSS3 (custom stylesheets, no CSS framework)
Front-end scripting	Vanilla JavaScript (Fetch API for all client–server communication)
QR code scanning	html5-qrcode (open-source, loaded via CDN)
QR code generation	api.qrserver.com on-the-fly QR image generation, keyed to each book_id
Back-end scripting	PHP with the MySQLi extension and prepared statements
Database	MySQL / MariaDB (relational)
Development environment	Local Apache/MySQL stack (XAMPP/WAMP-style), MySQL served on a non-default local port
Typography	Google Fonts – Roboto Slab, Segoe UI and Dancing Script

4. Results and Discussion

The implementation of the Smart E-Library Management System yielded highly positive results, successfully addressing the core inefficiencies identified in the traditional manual ledger system. The following sections evaluate the system's performance and data collected during the testing phase against the initial project objectives.

4.1. Achievement of Project Objectives

QR Code Integration for Book Processing: The library’s html5-qrcode integration successfully allowed students and library staff to scan books using standard device cameras. Tests showed a high success rate in decoding QR codes without interruption, completely eliminating manual data entry errors that were common in the previous system.

Role-based user interfaces: Dedicated dashboards for administrators and students were implemented and successfully tested. Security validations confirmed that unauthorized users (students) were strictly restricted from accessing administrative functions, thereby maintaining data integrity.

Live inventory and catalog management: The database effectively updates book statuses (available, reserved, borrowed) in real time. Students also reported that checking book availability before visiting the physical library significantly optimized their time and reduced unnecessary waiting.

Automated Fine Calculation: The background logic for calculating late fines based on due_date and return_date was accurately automated. Test cases involving overdue books, accurately calculated penalties by retrieving current fine rates from the system_settings table, greatly reducing the administrative workload for staff**.**

4.2.System Testing and Performance

Extensive black box testing was conducted to ensure system stability. The main focus was on limiting the combined borrowing limit. The system was tested by having a student attempt to borrow or reserve a third book while they already had two active records.

The system's cross-validation mechanism successfully blocked all such attempts, confirming the reliability of the underlying PHP logic. Furthermore, the system response time for database queries was kept to a minimum, ensuring a smooth and responsive user experience even with multiple people attempting to access it simultaneously.

4.3.Discussion and Evaluation

"The main point confirmed by these test results is that using a web-based system created using QR technology is more efficient than the traditional library system. When comparing it with the facts we studied in the second chapter (Literature Review), it was found that technologies such as RFID and IoT cost a lot of money, but this new system we introduced has successfully overcome that cost problem. The specialty here is that there is no need to purchase dedicated scanning hardware for this. Since it can be easily implemented using currently used smart phones and computers, it has been designed at a low cost and can be easily used by anyone. Overall, this has made the daily activities of the library more systematic, and has been able to provide a very practical and successful solution to the basic problem of our research by minimizing the use of human labor (Manual intervention)."

5. Conclusions and recommendations

5.1. Conclusion

Through this project, the "Smart E-Library Management System" was successfully designed and implemented, providing a permanent solution to the inefficiencies and weaknesses of the traditional library system. Building a robust database using PHP and MySQL technologies, and digitizing the issuance and receipt of books through HTML5-QRCode technology have made the daily operations of the library much easier.

In terms of security, the security of data has been well ensured by providing separate logins for students and administrators. Instead of technologies such as RFID, which incur high costs, the introduction of the QR code scanning facility through already used smart phones is a unique achievement of this project. Ultimately, this system has successfully achieved all the basic objectives of saving students' time and minimizing the workload of library staff.

5.2.Recommendations for Future Enhancements

Although this system is already operating successfully, I recommend that the following features be added to the system to further develop it in the future:

- SMS or Email Notifications: Adding the facility to automatically send an SMS or Email message to the system when the due date of the books borrowed by students is approaching or has expired.
- Online Payment Gateway: Facilitating the payment of fines for overdue books online (via cards) without visiting the library.
- Introducing a Mobile Application: In addition to the existing web system, creating a separate Mobile App for Android and iOS so that students can use it more easily.
- AI-based Recommendations: Including a feature that analyzes the reading history of the books a student has previously read and suggests new books that are suitable for them.